

Integrated Assessment of Stream Health in Four Creeks of the Upper Blue Mountains, NSW



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Acknowledgement

The Blue Mountains City lies on the Country of the Darug and Gundungurra Traditional Owners. We recognise their deep and ongoing connection to Ngurra (Country), which includes the land, water, culture, and spirit. We pay our respects to Elders past and present and acknowledge the strength and resilience of Aboriginal and Torres Strait Islander peoples.

The EPA Aboriginal artwork '*Nyuragil Barray Yawutung Warri*', meaning '*Spirited Country Journey Now*', was created by Gerard Black. It represents a journey across Country, from desert landscapes to mountainous regions.

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1. Introduction

The freshwater stream ecosystems are one of the most susceptible environments in the world, which are under pressure due to increasing urban development, shifts in land use and changes in climate patterns (Dudgeon et al., 2006; Allan, 2004). In addition to sustaining a variety of biological communities and ecosystem services including recreation and nutrient cycling, these systems are important in supplying clean water in domestic, agricultural and industrial activities (Vörösmarty et al., 2010; Yu et al., 2019). However, since 1970, the population of freshwater species reported to drop by an estimated 81% (WWF, 2016). This led to monitoring and assessment of stream health becoming a major priority in environmental management and policy.

Stream health is a complex issue as it depends on physical, chemical and biological processes. The quality parameters of water such as pH, dissolved oxygen, turbidity and nutrient content are directly related on aquatic organisms. Besides, the ecosystem stability and resilience depend on habitat properties such as riparian vegetation, channel structure and flow regime (Allan and Castillo, 2007; Choi et al., 2011). Biological communities like macroinvertebrates are highly responsive to environmental changes and are widely used as indicators of ecosystem health. (Bonada et al., 2006; Chessman, 2003).

To deal with this complexity, integrated assessment methods that involve the use of several indicators have become very common. In Australia, water quality is often assessed using the SIGNAL (Stream Invertebrate Grade Number Average Level) system, which evaluates how sensitive different macroinvertebrates are to pollution. In addition, local frameworks such as the Blue Mountains City Council (BMCC) waterway health rating system classify stream condition by combining biological data with habitat assessments.

The Blue Mountains area of New South Wales is a unique and ecologically important terrain typified by high altitude headwater streams, forested catchments and the growing urban and recreational pressures. The streams within this area are especially vulnerable to disruptions like stormwater runoff, erosion and other human activities related to tourism and residential development.

This study will assess stream health through an integrated method, which involves water quality assessment, macroinvertebrate-based SIGNAL analysis and riparian vegetation assessment in the Upper Blue Mountains. It will compare ecological conditions of sites to

determine important environmental pressures and give management recommendations on how to improve stream health in the area.

2. Study Area and Site Description

The research was carried out in the Upper Blue Mountains area of New South Wales (NSW), Australia, in four creek sites in and around Katoomba (Table 1). The area is part of a World Heritage-listed landscape, which is geologically sandy and forested, with ecologically valuable freshwater systems. The study area is a complex environmental environment where urbanization, tourism and natural ecosystems interact to produce different degrees of anthropogenic pressures on waterways.

Table 1: Summary of Study Sites and Site Characteristics

Site No.	Creek Name	Coordinates	Site Description
1	Maple Grove Park Creek	33.725132°S, 150.306549°E	A shallow, narrow creek located in a park–residential area (940 m elevation), with slightly turbid water following recent rainfall (3.2 mm). The site supports moderate riparian vegetation, including ferns, grasses, shrubs and Eucalypts, providing partial canopy cover. Evidence of human impact is present, including stormwater runoff, footpaths, litter and bank erosion. Change of open parkland to natural vegetation witnessed.
2	Upper Kedumba River	33.72399°S, 150.30395°E	Having changing conditions along the course. Upstream, open woodland over sewer pipe. Midstream, modified channel with concrete-covered sewer pipe and gabion baskets leading to pooling. Downstream, redwood plantation with shallow, rocky channel. The surrounding land use includes a car park, oval, and tourist facilities, indicating a moderate level of human influence.
3	Katoomba Creek	33°41'38.8"S, 150°18'43.0"E	Narrow, shallow creek of moderate flow and dense riparian vegetation with Lomandra, ferns, Banksia and Eucalyptus. Situated next to floodplain vegetation and

Site No.	Creek Name	Coordinates	Site Description
			residential area. Signs of plastic waste and possible urban runoff. There is vegetation density that varies along the stream, riffle and habitat diversity exist. The accessibility of urban areas affects the ecosystem and water quality.
4	Yosemite Creek	33.68485°S, 150.32647°E	The creek discharges into Minnehaha Falls and is influenced upstream by residential areas and waste management facilities. It has a native vegetated riparian zone but is located near a high-traffic walking track, with potential <i>E. coli</i> risks from domestic animals. A stormwater retention basin is also present upstream. Creek is small and shallow with small pools and small waterfalls.

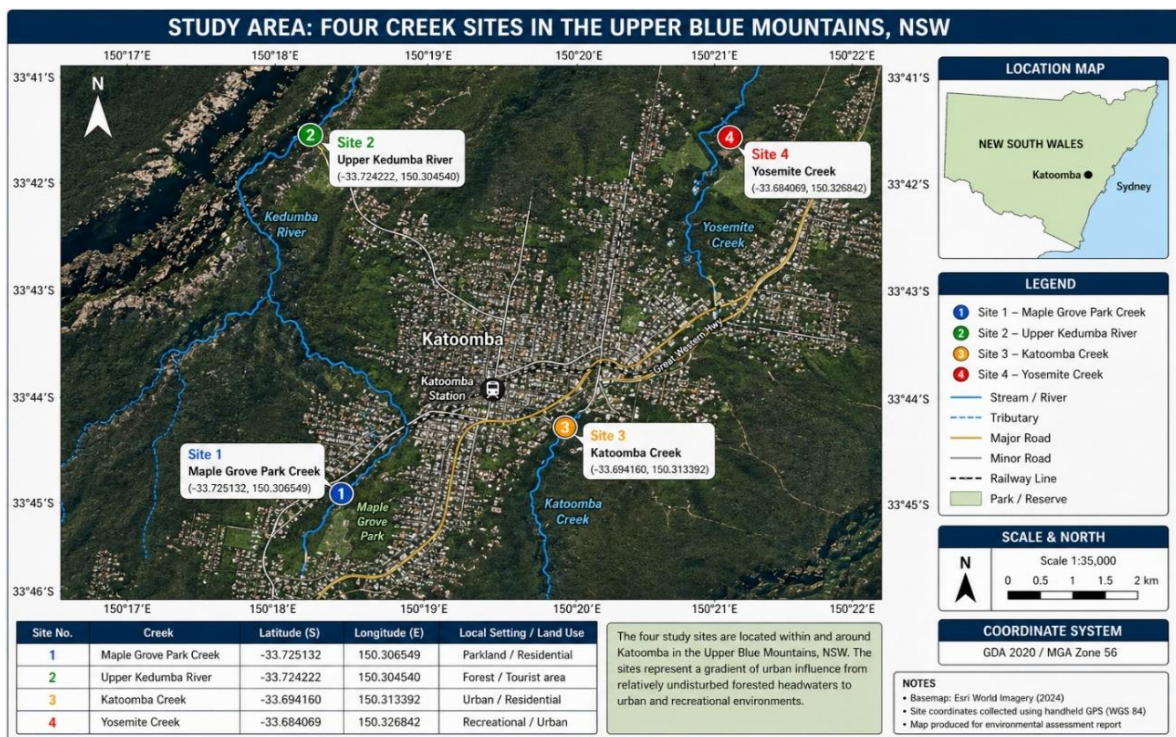


Figure 1: Location of the four creek study sites in the Upper Blue Mountains, NSW.

3. Methodology

3.1 Site Description and Environmental Assessment

A qualitative site assessment was conducted at each location to observe environmental conditions and potential impacts. Observations included, surrounding land use, weather conditions and recent rainfall history, channel morphology and flow characteristics, bank condition and evidence of erosion, visible pollution (e.g., litter, pipes, runoff pathways). Stream longitudinal profile was also measured using an inclinometer and tape measure.

Sketch maps and photographs were used to record site features. Attention was given to identifying sources of disturbance, stormwater drains, sewage infrastructure and human activities.

3.2 Water Quality Assessment

Water quality was measured in the field using a Hanna multi-parameter probe and standard methods. The following parameters were recorded:

- Electrical Conductivity (EC)
- Dissolved Oxygen (DO)
- pH
- Turbidity (NTU)
- Nitrate (mg/L)
- Phosphate (mg/L)
- Water Temperature
- E. Coli (CFU/100 mL)

Turbidity was estimated using the black disc method. Nitrate and Phosphate was measured using a spectrophotometer following standard procedures. All instruments were calibrated prior to measurement to ensure accuracy. Then measured values were compared with the Blue Mountains City Council (BMCC) trigger values for waterways above 500 m altitude.

Table 2: Desired water quality ranges for Blue Mountains waterways (>500 m altitude) (Blue Mountains City Council, 2018)

EC ($\mu\text{S/cm}$)	pH	DO (%) sat)	Turbidity (NTU)	Alkalinity (ppm CaCO_3)	Nitrate-N (mg/L)	Faecal Coliforms (CFU/100 mL)
< 65	4.37 – 7.24	63.28 – 100.38	< 12.25	< 10.9	< 0.22	< 20

3.3 Riparian Vegetation Assessment

Riparian vegetation was assessed at each site to evaluate habitat quality, bank stability and ecological condition. At each site, vegetation surveys were conducted along the stream banks using transect-based (5m*5m) observations. The following parameters were recorded:

- Canopy cover (%)
- Native canopy (%)
- Ground cover (%)
- Understorey cover (%)
- Native understorey (%)
- Number of vegetation layers
- Native ground cover (%)

Vegetation cover was estimated visually within the transect area. Riparian vegetation condition was assessed using a structured scoring framework evaluating habitat, vegetation cover, native species, debris, and regeneration features.

The assessment included:

- **Habitat structure:** canopy continuity, width and proximity to the stream
- **Vegetation cover:** canopy, understorey, ground cover, native components, and number of layers
- **Debris:** leaf litter, dead wood, and fallen logs
- **Regeneration features:** native species regeneration and presence of key vegetation types

Each parameter was assigned a score based on field observations and component scores were calculated as:

$$\text{Total Score} = (A + B + C) + (D + E + F + G) + (H + I + J) + (K + L + M + N + O) + (P + Q + R + S)$$

The maximum possible score was 50, with higher values indicating better riparian condition and ecological health.

3.4 Macroinvertebrate Sampling

Macroinvertebrates were sampled to assess biological condition and detect potential pollution impacts. Sampling was conducted using sweep nets. At each site, two samples were collected to ensure representative coverage.

Collected material was transferred to sorting trays. Macroinvertebrates were manually separated using forceps and pipettes and preserved in 70% ethanol for later identification in the laboratory.

Macroinvertebrate data were further analysed to assess stream health using a combined SIGNAL and BMCC (Blue Mountains City Council) waterway health assessment framework.

All collected macroinvertebrates were identified to family level and each family was assigned a SIGNAL grade based on its sensitivity to pollution. SIGNAL grades range from 1 (highly tolerant) to 10 (highly sensitive), with higher values indicating better water quality.

The SIGNAL score for each site was calculated as the average of all assigned SIGNAL grades, excluding non-scored (NS) taxa:

$$\text{SIGNAL Score} = \frac{\sum \text{SIGNAL grades}}{\text{Number of taxa (excluding NS)}}$$

In addition to the SIGNAL score, three key biological metrics were calculated:

- Total number of families (taxa richness)
- Number of EPT families (Ephemeroptera, Plecoptera, Trichoptera)
- Percentage of EPT taxa (% EPT)

These metrics provide an indication of biodiversity and the presence of pollution-sensitive organisms.

Each metric (SIGNAL score, taxa richness, EPT richness, and % EPT) was then assigned a factor score ranging from 0 to 5 using predefined classification bands. Higher factor scores indicate better ecological condition. (Table 3)

Table 3: Classification of Water Quality Based on SIGNAL Score (Blue Mountains City Council, 2018)

Site SIGNAL Score	Water Quality Rating
> 6.0	Healthy Habitat
5.0 – 6.0	Mild Pollution
4.0 – 5.0	Moderate Pollution
< 4.0	Severe Pollution

An average factor score was calculated for each site:

$$\text{Average Factor Score} = \frac{\text{Sum of all factor scores}}{\text{Number of metrics}}$$

The average factor score was then used to assign an overall waterway health rating (Very Poor, Poor, Fair, Good, or Excellent) based on the BMCC classification system. (Table 4)

This integrated approach allows for a comprehensive assessment of stream health by combining biological sensitivity (SIGNAL), biodiversity (taxa richness), and presence of sensitive taxa (EPT).

Table 4: Allocating Waterway Health Ratings (Blue Mountains City Council, 2018)

Average Factor Score	Waterway Health Rating
0.00 – 0.99	Very Poor
1.00 – 1.99	Poor
2.00 – 2.99	Fair
3.00 – 3.99	Good
4.00 – 5.00	Excellent

4.Result

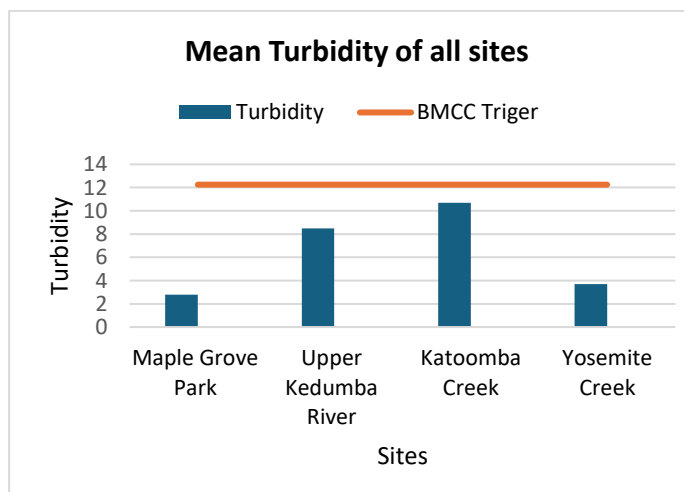
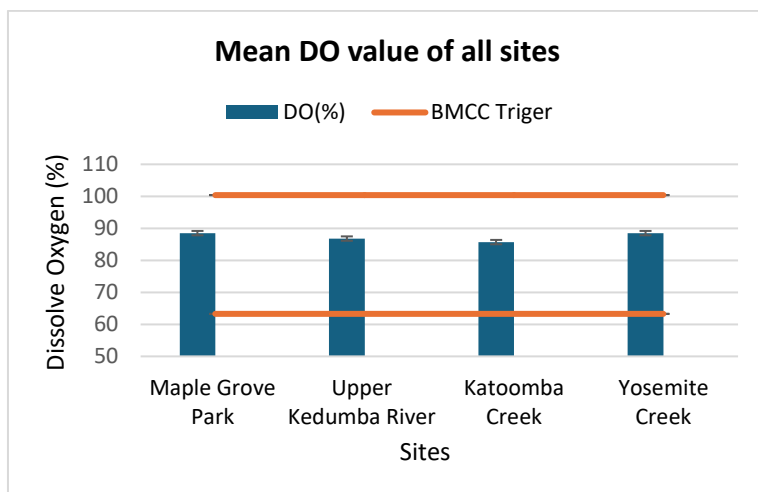
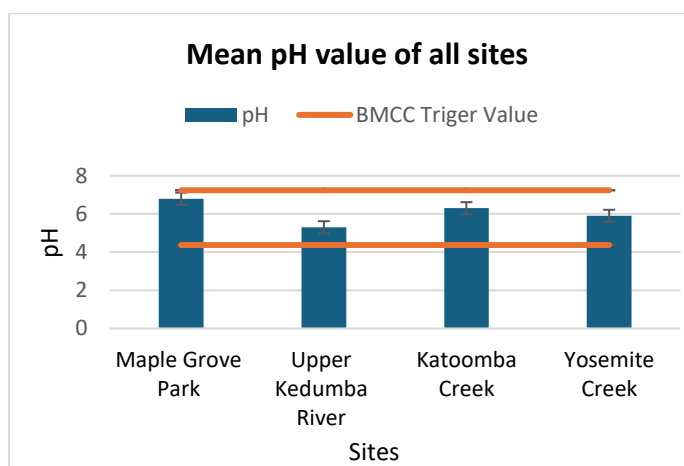
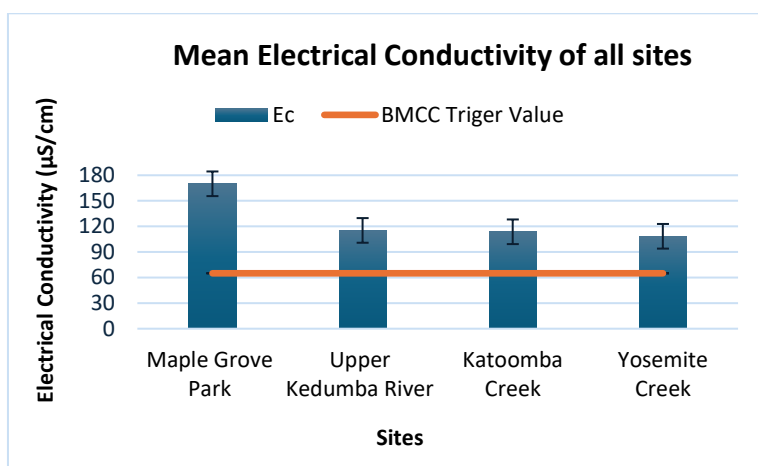
Table 5 shows, the temperature was between 12.4°C and 14.4°C with little difference in temperature among sites. The electrical conductivity (EC) was greater than the guideline (<65 µS/cm) at each site, with the highest value at Maple Grove Park (170.0 µS/cm). The level of dissolved oxygen was in the acceptable range (63.28-100.38) in all sites and pH was within the guideline (4.37-7.24). Turbidity was lower than the threshold (<12.25 NTU) in all locations with the highest at Katoomba Creek (10.6 NTU). Nitrate and phosphorus levels were more than guideline values at all locations with the highest nitrate levels in Katoomba Creek (2.2mg/L) and highest phosphorus in Maple Grove Park (0.4mg/L).

Table 5: Water Quality Parameter of all sites

Parameter	BMCC Triger Value	Maple Grove Park	Upper Kedumba River	Katoomba Creek	Yosemite Creek
Temperature (°C)	-----	14.2 ± 0.230072	13.5 ± 0.445234	12.4 ± 0.041633	14.4 ± 0.100167
Electrical conductivity (µS/cm)	< 65	170.0±4.58257	115.3±0.578315	113.7±0.57735	108.4±2.499927
Dissolve Oxygen (%)	63.28 – 100.38	88.5 ± 0.820752	86.8 ± 2.401506	85.7 ± 0.856524	88.5 ± 3.476569
pH	4.37 – 7.24	6.8±0.113725	5.3±0.132791	6.3 ± 0.205264	5.9 ± 0.407226
Turbidity (NTU)	< 12.25	2.8 ± 0.111355	8.5 ± 3.821012	10.6 ± 3.373208	3.7±0.417612
Nitrate (mg/L)	0.22	1.6±2.72E-16	0.6 ± 0.520416	2.2 ± 0.05	1.4 ± 0.284312

Phosphorous (mg/L)	0.007	0.4 ± 0.2326119	0.2 ± 0.095393	0.1 ± 0.14580237	0.2 ± 0.05
E.Coli (CFU)	< 20	63.7 ± 11.15049	27.0 ± 18.73499	17.3 ± 8.386497	6.7 ± 3.05505

i. Water Quality Comparison of all sites



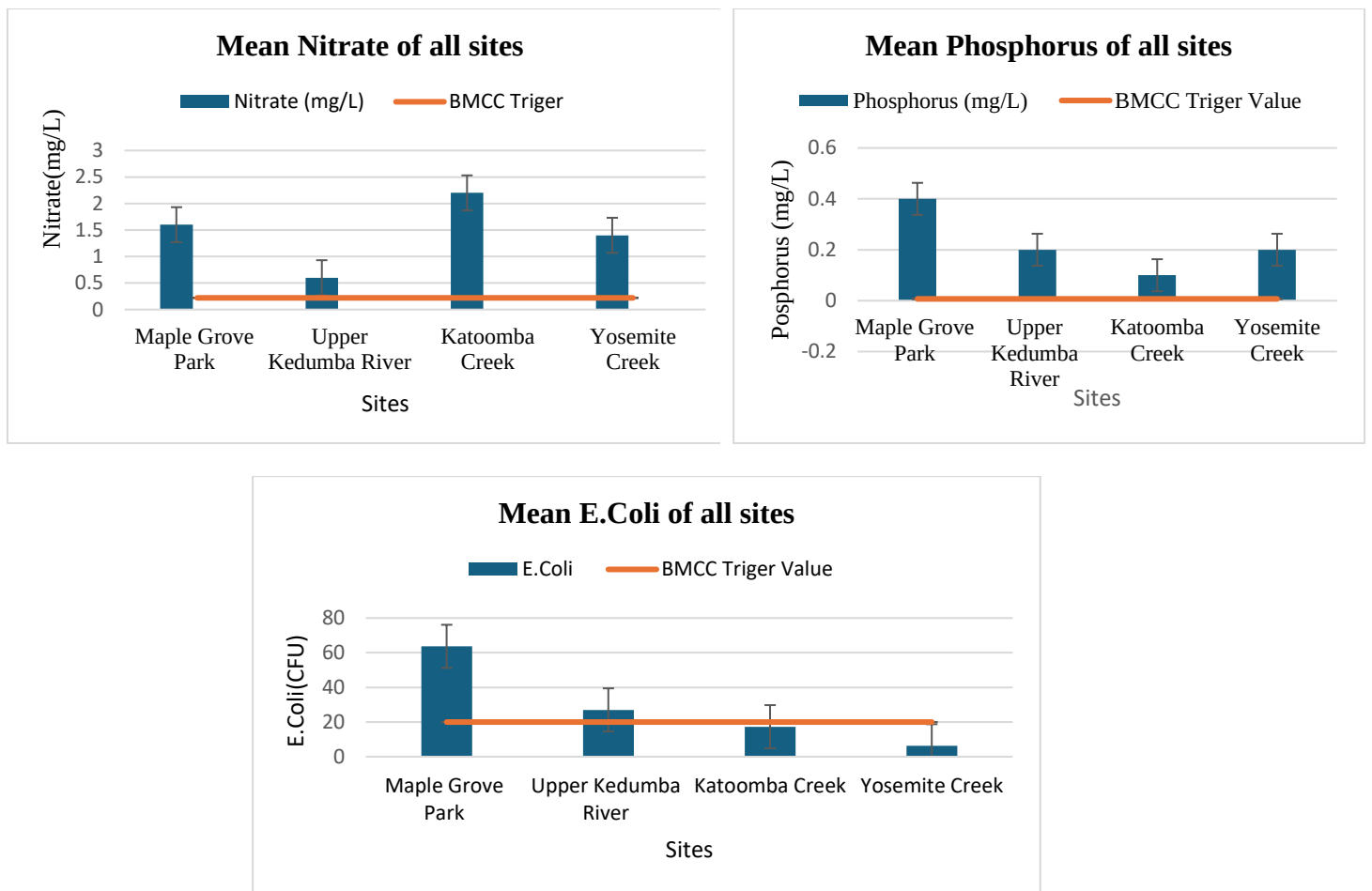


Fig 2: Water Quality Parameter Comparison of all sites

The graphs (Fig 2) indicate that the water quality varies between the four sites. The values of pH and dissolved oxygen are within the BMCC acceptable range at all the sites. But, the electrical conductivity values, nitrate and phosphorus appear to be far above the allowed values in all the sites. In this case, the values are found to be higher at Maple Grove Park especially with EC and E. coli and lower at Yosemite Creek.

ii. Riparian Vegetation

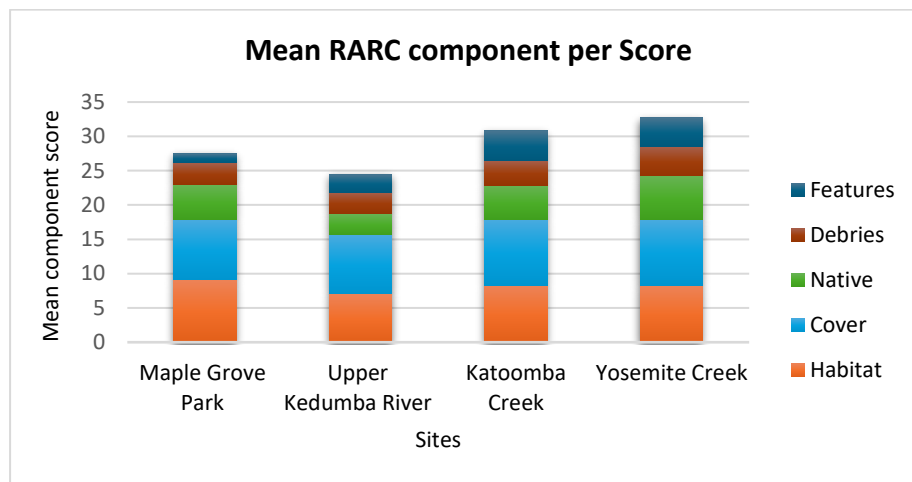


Figure 3: Comparison of Mean Riparian Vegetation Assessment Components Across Four Creek Sites

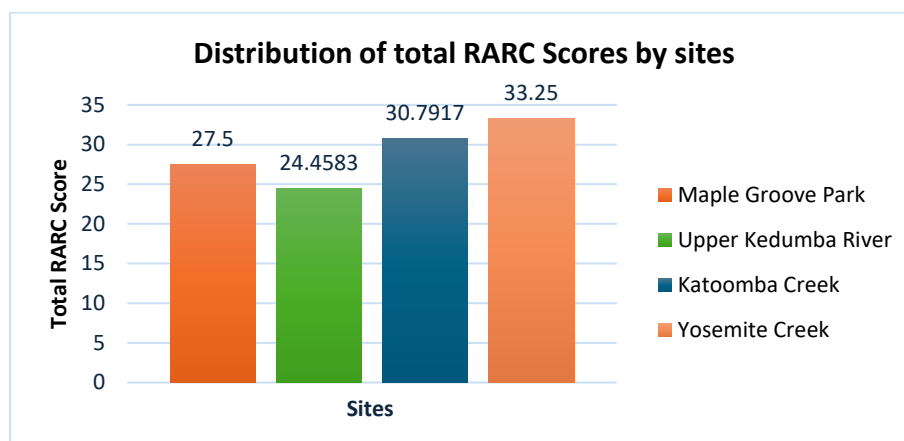


Figure 4: Distribution of Total RARC Scores by sites

Figure 3 and figure 4 shows, the component scores between sites in the riparian vegetation graphs vary to some extent. Yosemite Creek has the largest total RARC score. On the contrary, the lowest total scores are in Upper Kedumba River. The average riparian conditions are in the middle of Maple Grove Park and Katoomba Creek.

iii. Macroinvertebrate of all site

Table 6: Comparison of Waterway Health Ratings Based on BMCC Factor Scores Across Study Sites

Name of the Site	Average of factor score (compared with urban)	Waterway health rating	Average of factor score (compared with the reference)	Waterway health rating
Maple Grove	0.41	Very poor	0.29	Very poor
Kedumba River	1.29	Poor	0.83	Very Poor
Katoomba Creek	1.45	poor	0.95	Very poor
Yosemite Creek	2.08	Fair	0.958	Very poor

Table 6 shows that the four sites had different waterway health ratings in terms of BMCC factor scores. Compared to urban reference conditions, Maple Grove was considered as Very Poor (0.41), Upper Kedumba River and Katoomba Creek as Poor (1.29 and 1.45) and Yosemite Creek as Fair (2.08). Comparatively, in comparison to reference (natural) conditions, all sites scored Very Poor with the range of 0.29 to 0.96.

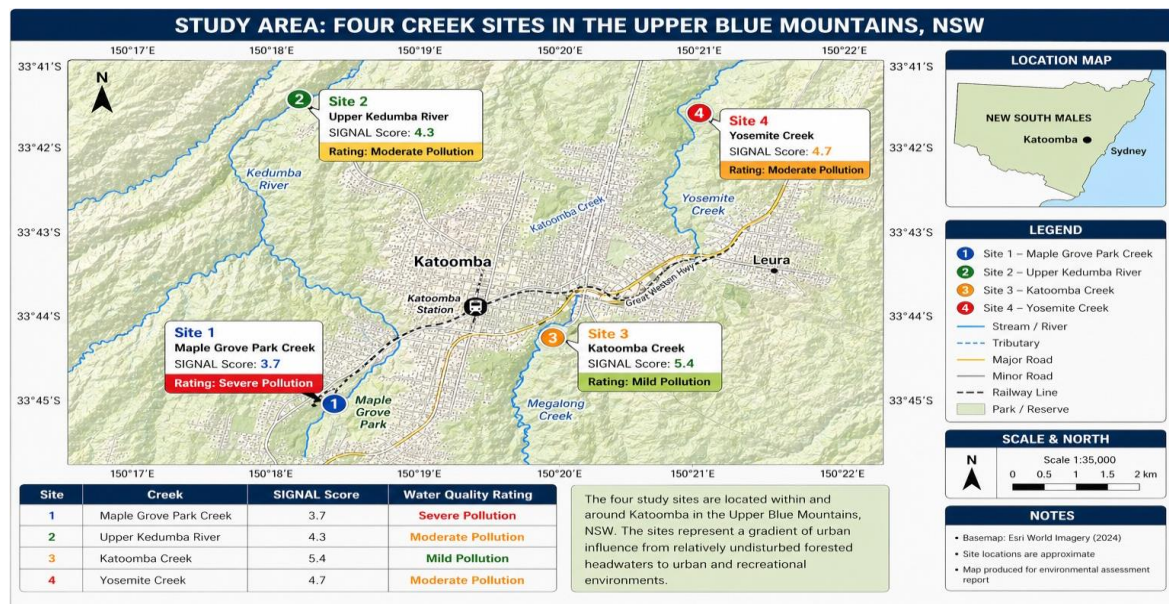


Figure 5: Map Showing Spatial Variation in Water Quality Ratings (SIGNAL Scores) Across Four Creek Sites in the Upper Blue Mountains

Figure 5 indicates the differences in the health of the streams at the sites according to SIGNAL scores. The lowest score (3.7, indicating severe pollution) was recorded at Maple Grove Park, while the highest score (5.4, indicating mild pollution) was observed at Katoomba Creek.

5. Discussion

This study demonstrates the variability in stream health across the four sites, which relate to variations in catchment land use, human pressure and riparian zone status.

In the water quality analysis, the pH and dissolved oxygen (DO) levels at all sites were within the BMCC recommended levels, suggesting that the basic chemical parameters were suitable for supporting aquatic life. But other parameters including electrical conductivity (EC), nitrate, and phosphorus were considerably different from the standard values, implying environmental disturbance.

All sites had significantly higher electrical conductivity values than the maximum permissible level ($65 \mu\text{S}/\text{cm}$) with the highest at Maple Grove Park. This is an indication of an increased concentration of dissolved ions, likely due to urban runoff, stormwater runoff and potentially sewage. This finding is in agreement with the site description listing stormwater inputs, walking tracks and residential influence.

All sites showed nitrate values above the BMCC standard ($0.22 \text{ mg}/\text{L}$) and the highest concentration was found in Katoomba Creek. This can be explained by the inputs of residential run-off and organic wastes. Concentrations of phosphorus were also much higher than the standard, suggesting potential for nutrient enrichment and development of eutrophication. This indicates that all sites are impacted by nutrient pollution.

Turbidity levels were generally within acceptable ranges; however, the elevated turbidity observed in the Upper Kedumba River and Katoomba Creek may be attributed to channel modification, bank erosion, and human disturbance.

There were large variations in microbial contamination between the sites. *E. coli* levels were much higher than acceptable limits ($< 20 \text{ CFU}/100 \text{ mL}$) at Maple Grove Park, suggesting potential faecal contamination due to stormwater, pets or human presence. On the other hand, Yosemite Creek had lower *E. coli* values, indicating higher water quality.

The riparian assessment (RARC scores) showed moderate ecological health in riparian vegetation at all locations. Yosemite Creek scored the highest (33.25), reflecting the presence of well-developed vegetation structure, improved habitat and more abundance of native species. Lower Kedumba River scored the lowest, which could be due to channelisation, development, and loss of natural vegetation.

This water quality trend was supported by macroinvertebrate analysis. The majority of sites were rated as "Very Poor" and "Poor", which suggests conditions of ecosystem stress and pollution. The highest impact is observed in Maple Grove Park, which also has high nutrient concentrations, EC, and E. coli loads. Yosemite Creek generally remained in a better condition (rated as "Fair" by comparison with urban sites), probably because of the relatively better vegetation cover and disturbance level.

In general, the findings indicated that human factors, including urban development, tourism, stormwater runoff and infrastructure interventions are the key drivers of stream health in the study area. Impacts from urban and riparian zone disturbances, including lack of vegetation cover, were evident in sites with lower EPT and higher impairment ratings, such as Yosemite Creek, suggesting the need for better riparian management.

6. Conclusion

This work reveals variability in stream health among four sites, which can be associated with human use and land-use patterns. The pH and dissolved oxygen were in acceptable range, but the other water quality parameters - electrical conductivity, nitrate and phosphorus - went above the standard levels, suggesting pollution, particularly from urban and stormwater runoff.

Of the four sites, it was found that Maple Grove Park had the poorest health with increased nutrient loads and microbial growth, while Yosemite Creek was relatively better due to healthy riparian zones and less disturbance. This was also reflected in the macroinvertebrate data, with most sites in poor health.

The overall results reveal human activities are affecting these streams, and enhancing the health of their vegetation and reducing pollution are critical for improving stream quality.

7. Management Recommendations

In Site 1 (Maple Grove Park Creek) we observed a small, slightly turbid creek with significant vegetation, evidence of stormwater runoff, garbage and bank erosion. There were some fallen/broken trees (timber) across the creek, it should be repositioned if it blocks the water flow. For better stormwater and bank erosion management, vegetation should be planted along eroded banks and minimise human impacts through improved pathway design and litter removal.

In Site 2 (Upper Kedumba River Observed) we observed channelised, exposed and covered sewer lines, gabions creating pools, affected by tourism and urban development. Need to maintain sewer lines, remove blockages and allow flow to return to natural conditions, control runoff from carparks and other infrastructure. Establish buffer vegetation zones.

In site 3 (Katoomba Creek) we observed healthy riparian vegetation but affected by urbanisation and plastics, moderate flow, habitat diversity. Urban runoff should be regulated, with traps installed and community education promoted to reduce littering and protect the environment. In addition, a strong riparian vegetation buffer should be maintained.

In site 4 (Yosemite Creek) we observed good riparian vegetation but were impacted by a stormwater basin, adjacent tip and high human (and pet) activity. Need to manage stormwater basins, reduce human and pet impacts and track potential pollution sources. Maintain and protect existing riparian vegetation.

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Appendix

BMCC waterway health rating System for Maple Grove Park

Table 1: Maple Grove Park waterway health rating

Factor	Metrics	Comparison with Urban	Rating (comparison with reference)
Signal Score	3.7	0	0
Total Number of Types (Family richness incl. NS bugs)	20	1	1
Number of EPT texa	1	1	1
% EPT of total no. of Families	5	1	1
Average	n/a	0.41	0.29
Water health rating	n/a	Very poor	Very poor

BMCC waterway health rating System for Kedumba River

Table 2: Upper Kedumba River waterway health rating

Factor	Metrics	Comparison with Urban	Rating (comparison with reference)
Signal Score	4.3	0	0
Total Number of Types (Family richness incl. NS bugs)	48	2	1
Number of EPT texa	10	5	3
% EPT of total no. of Families	20.83	2	1
Average	n/a	1.29	0.83
Water health rating	n/a	Poor	Very poor

BMCC waterway health rating System for Katoomba Creek

Table 3: Katoomba Creek waterway health rating

Factor	Metrics	Comparison with Urban	Rating (comparison with reference)
Signal Score	5.4	5	1
Total Number of Types (Family richness incl. NS bugs)	27	1	1
Number of EPT texa	7	4	2
% EPT of total no. of Families	25.92	2	1
Average	n/a	1.45	0.95
Water health rating	n/a	Poor	Very poor

BMCC waterway health rating System for Yosemite Creek

Table 4: Yosemite Creek waterway health rating

Factor	Metrics	Comparison with Urban	Rating (comparison with reference)
Signal Score	4.7	1	0
Total Number of Types (Family richness incl. NS bugs)	66	3	1
Number of EPT taxa	22	5	4
% EPT of total no. of Families	33.33	2	1
Average	n/a	2.08	0.958
Water health rating	n/a	Fair	Very Poor

Riparian Vegetation

Table 5: Riparian vegetation of different sites

Name of the site	Mapple Grove Park	Upper Kedumba River	Katoomba Creek	Yosemite Creek
Habitat	9.083333	7.041667	8.33333	8.33333
Cover	8.75	8.625	9.541667	9.583333
Native	5.16667	3.08333	5	6.33333
Debries	3.125	3.125	3.625	4.25
Features	1.375	2.58333	4.291667	4.25
Total RARC Score	27.5	24.4583	30.7917	33.25

Site Sketch and Photos



Figure 6: Maple Grove Creek (site 1)

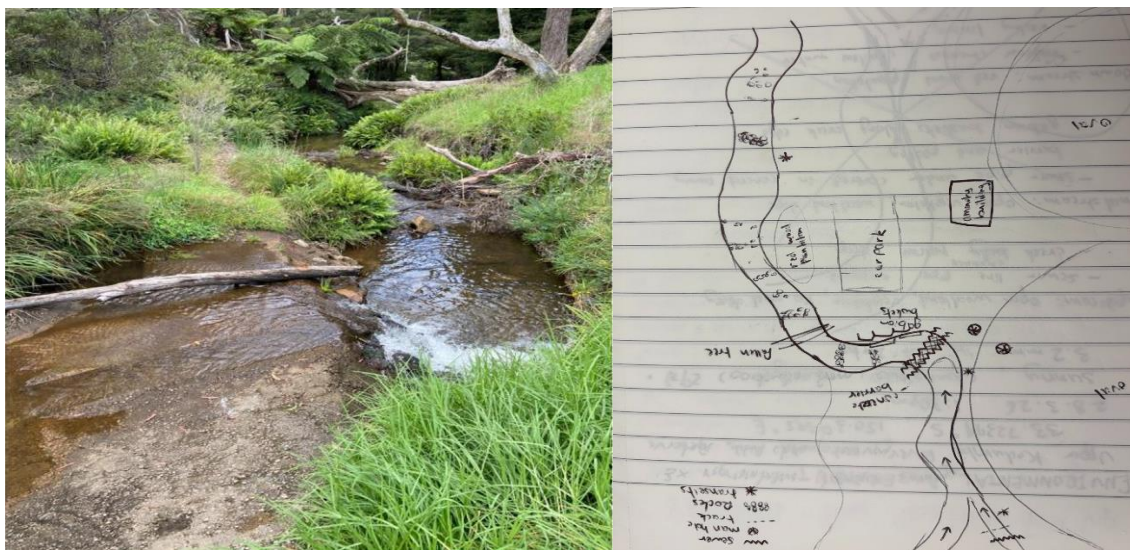


Figure 7: Upper Kedumba Rive (site 2)

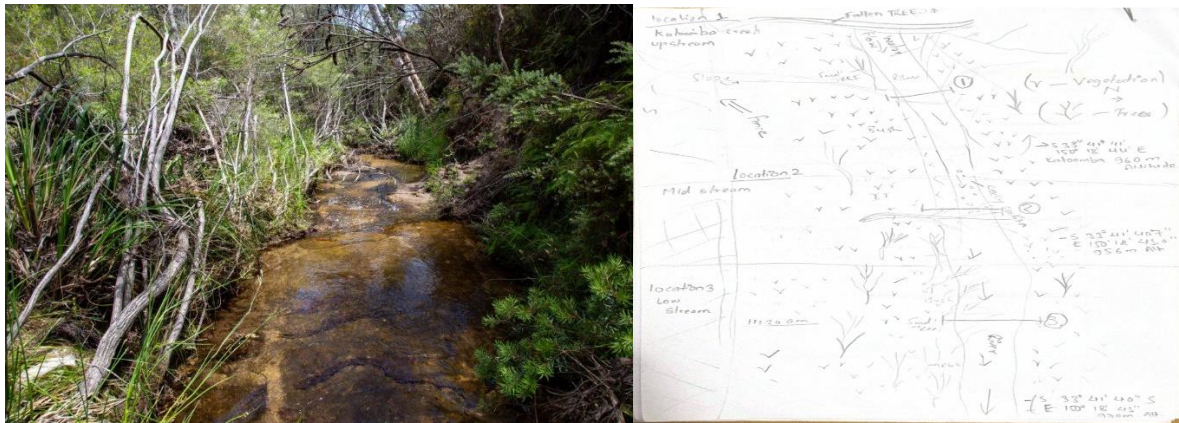


Figure 8: Katoomba Creek (site 3)

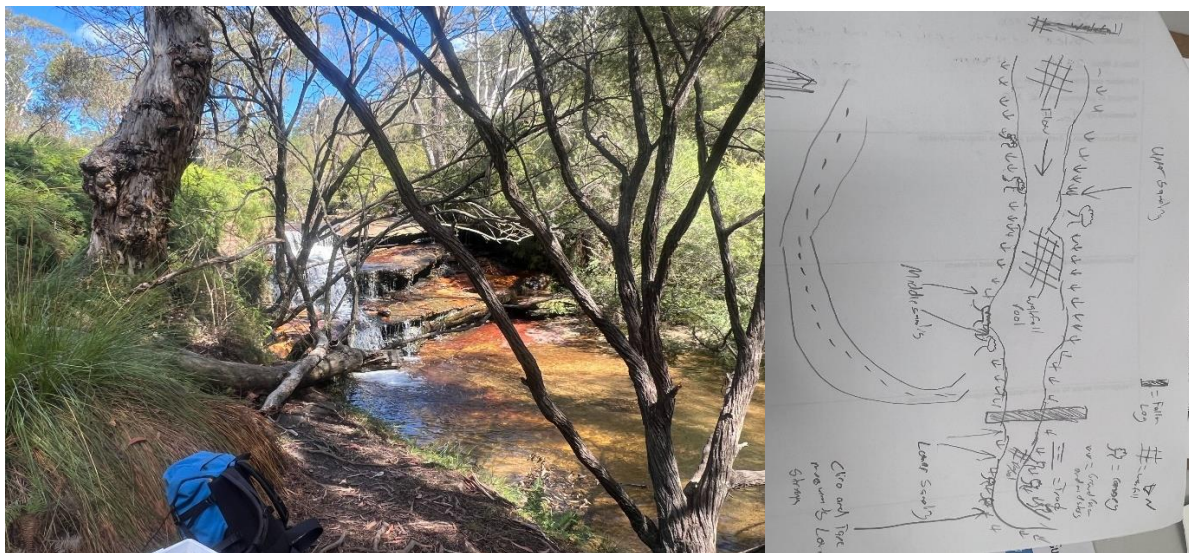


Figure 9: Yosemite Creek (site 4)